

Decent Coaching Classes
IX CBSE
Mind Your Math - IV

Subject: Mathematics Chapters: 2, 4, 5, 7

Marks: 30

Time: 1 hour

Instructions:

- i. This question paper has four sections: A, B, C and D.
- ii. Section A has 5 MCQs of 1 mark each.
- iii. Section B has 4 questions of 2 marks each.
- iv. Section C has 3 questions of 3 marks each. All contain an internal choice as indicated.
- v. Section D has 2 questions of 4 marks each.
- vi. Draw neat, labelled figures/graphs wherever necessary.

Section A – Multiple Choice Questions

(5)

1. The degree of the polynomial $5x - 3$ is:
(A) 0 (B) 1 (C) 2 (D) 3
2. $a^3 - b^3 =$ _____
(A) $(a - b)(a^2 + ab + b^2)$
(B) $(a + b)(a^2 - ab + b^2)$
(C) $(a - b)(a^2 - ab + b^2)$
(D) $(a - b)^3$
3. The perpendicular drawn from the centre of a circle to a chord _____ the chord.
(A) trisects (B) bisects (C) doubles (D) none of these
4. The total number of outcomes when a coin is tossed twice is:
(A) 2 (B) 3 (C) 4 (D) 6
5. The probability of an impossible event is:
(A) 1 (B) 0.5 (C) 0 (D) undefined

Section B – Very Short Answer Questions

(8)

1. Find the value of the polynomial $p(y) = 3y^2 - 2y + 7$ at $y = -1$.
2. A chord of a circle is 16 cm long and is at a distance of 6 cm from the centre. Find the radius of the circle.
3. Two coins are tossed together 60 times, and the outcome “two heads” appeared 15 times. Find the experimental probability of getting two heads.
4. Using a suitable identity, factorise: $9x^2 + 24xy + 16y^2$

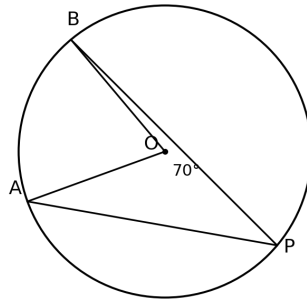
OR

Using a suitable identity, find the value of 105×95 .

Section C – Short Answer Questions

(9)

1. In the figure given below, O is the centre of the circle and the chord AB subtends an angle of 70° at the centre. Find the measure of $\angle APB$, where P is a point on the major arc. Give reason for your answer.



2. A bag contains 5 red, 7 blue and 8 green marbles. A marble is drawn at random from the bag. Find the probability that the marble drawn is (i) red, (ii) not blue, (iii) either red or green.

3. Expand using a suitable identity: $(2p - 3q + 4)^2$

OR

Factorise using a suitable identity: $x^3 - 64y^3$

Section D – Long Answer Questions

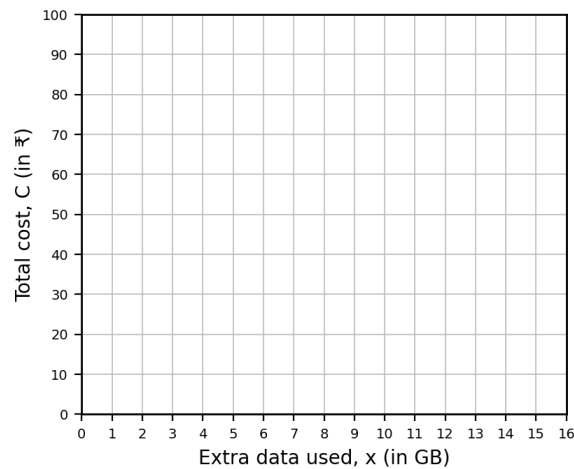
(8)

1. A mobile recharge plan charges a fixed amount of ₹50, plus ₹2 per extra GB of data used beyond the free limit. The total cost C (in ₹) for x extra GB used is given by $C(x) = 2x + 50$.

(i) Find the total cost C when $x = 10$ GB.

(ii) Find the value of x if the total cost is ₹80.

(iii) On the grid given below, plot the graph of $C = 2x + 50$ for $x = 0, 5, 10$ and 15 . Use your graph to estimate the total cost when $x = 12$ GB.



2. Draw $\triangle ABC$ with $AB = 6$ cm, $\angle A = 60^\circ$ and $\angle B = 75^\circ$. Draw the circumcircle of $\triangle ABC$. State whether the circumcentre lies inside, on, or outside the triangle, giving reason for your answer.

— End of Paper —